

Chapter 42 — Bladder Anomalies in Children — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Fetal bladder empties every 15–20 min** → re-scan before calling it absent.
- **First-trimester megacystis:** ≤ 6 mm normal; ≥ 8 mm abnormal.
- **Later:** SL (sagittal length, mm) = GA (weeks) – 5 (± 7); $> GA+2$ enlarged, $> GA+12$ megacystis.
- **7–15 mm** → most resolve; > 15 mm → none, all obstructive.
- **Early (< 18 wk) + resolved < 23 wk** → no postnatal surgery.
- **Megacystis commonest cause:** PUV (posterior urethral valves); M:F 8:1.
- **MMIHS** (megacystis–microcolon–intestinal hypoperistalsis): female, normal/ $\uparrow$ fluid, acontractile bladder, ACTG2.
- **Bladder agenesis:** survival only with ectopic ureters.
- **Symptomatic urachal remnant** → excise with bladder cuff.
- **Nephrogenic adenoma recurs 80%** → surveillance cystoscopy.

CONGENITAL BLADDER ANOMALIES

- **Embryology:** cloaca divided wk 4–6; urine from wk 10; bladder descends, urachus forms months 4–5.
- **Urachus** → median umbilical ligament.
- **Female urethra:** entirely pelvic urogenital sinus.
- **Fetal bladder:** wall ≤ 3 mm; diameter first trimester ≤ 6 –8 mm.
- **Visible:** ~half at 10 wk, almost all by 13 wk; transvaginal > transabdominal.
- **Amniotic fluid** = fetal urine only from 16 WGA (weeks' gestational age).
- **Fetal sex:** reliable after 14 wk; use testes, not phallus.
- **Cord:** 2 arteries + 1 vein, no fluid-filled urachus.
- **Agenesis:** ~1:600,000, female majority; ectopia into müllerian structures (F) / rectum (M).
- **Agenesis rescue:** ureterosigmoidostomy or stoma.
- **Exstrophy vs agenesis:** exstrophy = abdominal-wall template + normal fluid.
- **Covered cloacal exstrophy:** omphalocele + diastasis with bladder filling.
- **Hypoplasia:** epispadias, urogenital sinus, bilateral renal dysplasia, ectopia → augmentation often.
- **Megacystis cut-offs:** $< 10\%$ CRL (crown–rump length) normal, $\geq 13\%$ abnormal; others use > 6 or ≥ 7 mm.
- **After first trimester:** no emptying in 45 min = enlarged.
- **Aneuploidy:** ~one-fifth, mainly trisomy 13/18 → uniformly poor outcome.
- **Early megacystis ≤ 12 mm** → likely resolves.
- **Late (> 18 wk) + resolution > 23 wk** → most still need surgery.
- **Persistent megacystis + oligohydramnios** → LUTO (lower urinary tract obstruction).
- **Table 42.1 (pooled):** PUV ~50%, atresia/stenosis 7%, PBS (prune-belly) 4%, MMIHS 1%, cloacal 0.7%, undefined 36.5%.
- **MMIHS survival:** ~20% → ~55% recently; TPN (total parenteral nutrition) complications commonest death.
- **MMIHS gold standard:** exclude Hirschsprung by ganglion cells on colon biopsy.
- **MMIHS first-line:** CIC (clean intermittent catheterisation) / vesicostomy; lifelong CIC.
- **Duplication:** commonest complete sagittal; genital dup. $\leq 90\%$, lower GI $\leq 42\%$.
- **Sagittal > coronal** for non-urological anomalies.
- **Bladder without urethra** → ipsilateral renal dysplasia.
- **Operate if:** complete duplication → both sphincters competent: join urethras; one incompetent: close neck, excise urethra.

- **Don't operate:** incomplete duplication draining well; asymptomatic.
- **Diverticula gold standard:** VCUG (voiding cystourethrography).
- **Commonest congenital:** paraureteral.
- **Hutch:** primary = smooth, single, intermittent, no obstruction; secondary = trabeculated, multiple, constant, obstruction.
- **Operate if:** primary paraureteral + VUR (vesicoureteral reflux) → excise.
- **VUR resolution:** females > males (higher male voiding pressure).
- **Syndromic diverticula:** Ehlers–Danlos, Williams, Menkes; lax wall; can exceed bladder.
- **Operate if:** syndromic + UTI (urinary tract infection)/stones → resect; postop urodynamics.
- **Urachal cyst:** distal; adults > children; *S. aureus* commonest.
- **Patent urachus:** in-utero obstruction $\leq 14\%$ only.
- **Diagnosis:** patent urachus → ultrasound / fistulogram / VCUG; sinus → sinugram; cyst → ultrasound.
- **Infected remnant:** drain + antibiotics → excise.
- **Don't operate:** asymptomatic incidental remnant (trend; discuss with family); wide-necked urachal diverticulum.
- **Laparoscopic excision:** feasible < 6 months.

ACQUIRED BLADDER ANOMALIES IN CHILDREN

- **Acquired diverticulum:** treat obstruction first; remodelling may avoid excision.
- **Operate if:** still symptomatic → excise $\pm$ ipsilateral reimplant.
- **Waldeyer sheath damage** → UVJ (ureterovesical junction) failure → reflux/obstruction.
- **Haemangioma:** Klippel–Trénaunay, Sturge–Weber; solitary < 3 cm; gross haematuria.
- **Haemangioma treatment:** small → fulguration / Nd:YAG (neodymium:YAG) laser; large → partial cystectomy.
- **Fibroepithelial polyp:** 2nd commonest tumour after rhabdomyosarcoma; endoscopic resection.
- **Nephrogenic adenoma:** F:M 5:1; recurrence 80%, latency ~4 yr; distal tubular origin.
- **Cystitis cystica:** almost only female → long-term antibiotic prophylaxis.
- **Eosinophilic cystitis:** male; biopsy mandatory; TUR (transurethral resection) + steroids/antihistamines/antibiotics.
- **Don't operate:** eosinophilic cystitis in neonates — may self-limit.
- **Haemorrhagic cystitis:** cyclophosphamide/ifosfamide; $\uparrow$ with BK, CMV (cytomegalovirus), adenovirus; prevent with mesna.
- **Haemorrhagic cystitis adjunct:** oral pentosan polysulfate; avoid catheters.

CLASSIC TRAPS

- Trap: bladder not seen = agenesis — **Actually:** re-scan after 15–20 min.
- Trap: megacystis = PUV — **Actually:** over one-third undiagnosed.

- Trap: female megacystis → PUV — **Actually: MMIHS, cloacal, atresia.**
- Trap: prenatal resolution reassuring — **Actually: late resolution still needs surgery.**
- Trap: megacystis reflux from obstruction — **Actually: reflux can cause dilation.**
- Trap: early VCUG in MMIHS — **Actually: VUR uncommon, low yield.**
- Trap: paraureteral diverticulum bars Deflux — **Actually: injection still possible.**
- Trap: excise acquired diverticulum first — **Actually: fix outlet first.**
- Trap: sinugram separates urachal sinus/omphalomesenteric — **Actually: only histology.**
- Trap: eosinophilic cystitis looks typical — **Actually: no typical appearance, biopsy.**